Cochlear Implant in Far Advanced Otosclerosis

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Carlos Curet
National University of Córdoba

Results: VSR, WC, and age were positively associated with PTA score. But in men, VSR showed a positive association with borderline significance with PTA score. With backward stepwise multivariate linear regression analysis, VSR, but not WC, showed significant positive association with PTA score in females. When age, systemic disease, and other variables were adjusted using logistic regression analysis, WC and VSR showed significant positive correlation with hearing loss in males. However in women, VSR, but not WC, showed a significant positive correlation with hearing loss.

Conclusion: The VSR is an independent risk factor for ARHL. The VSR which is measured by CT is a better predictor of the risk of ARHI than is BMI or WC, especially for women.

Otology/Neurotology

Cholesteatoma: Long-term Follow-up in Romanian Patients

Marcel Cosgarea, MD (presenter); Violeta Necula, MD; Alma Maniu

Objective: Cholesteatoma is an abnormal destructive growth of squamous epithelium in middle ear and mastoid. The presence of cholesteatoma requires surgical intervention and the main goal is to achieve a safe, dry ear and to improve hearing. The extension of surgery depends on the size of the cholesteatoma.

Method: A total of 875 patients were treated for cholesteatoma in our department between 2001 and 2010, 213 pediatric patients. We performed both open and closed and technique, depending on disease extension, bone erosion, and complications, followed by second-look surgery in most of the closed-technique cases.

Results: The closed technique was the treatment of choice in 69.2% cases, most of them uncomplicated cholesteatomas. In 30.6% we performed open technique for different reasons. Second look surgery was done in 57.3% of cases after 1.5 or 2 years. A total of 27.1% of patients were lost to the follow-up program after the first years. Residual cholesteatomas were noticed in 17% of the cases and the recurrent cholesteatomas were operated in 32% of the cases. After surgery the air-bone gape improved in 67.1% of the cases because of tympanoplasty with autologous graft or different types of prosthesis. In some cases we used the BAHA prosthesis.

Conclusion: Cholesteatoma is a serious condition which needs surgical treatment. Second-look control in closed technique should be done after one-one and a half year. Closed technique is recommended in order to restore the function of the ear which is very important for a proper development and a good quality of life.

Otology/Neurotology

Cochlear Implant Outcomes of Cases with Temporal Bone Malformations

Hiroshi Yamazaki (presenter); Sho Koyasu; Saburo Moroto; Rinko Yamamoto; Tomoko Yamazaki; Keizo Fujiwara; Yasushi Naito, MD

Objective: 1) To reveal the relationship between cochlear implant (CI) outcomes and the types of temporal bone malformation.2) To establish a high-resolution CT (HRCT)–based classification of temporal bone malformations for predicting CI outcomes.

Method: We retrospectively evaluated 23 deaf children with cochleovestibular and/or internal auditory canal (IAC) malformations who underwent cochlear implantation from 2004 to 2010 at our hospital. Their temporal bone malformations were assessed by HRCT and their CI outcomes were evaluated by speech discrimination scores and category of auditory performance scores.

Results: We classified cochleovestibular and IAC malformations into 3 groups by 1) presence of a bony modiolus in the cochlea and 2) patency of cochlear nerve canal (CNC) and IAC. Eleven cases with a bony modiolus and patent CNC/IAC (Group 1) showed excellent CI outcomes and could communicate without visual language, while all of 6 cases with atresia of CNC or narrow IAC (Group 3) needed lipreading or sign language for understanding common phrases, even if their cochleae were well developed. The remaining 6 cases with a cystic cochlea and patent CNC/IAC (Group 2) showed varying CI outcomes depending on maturity of the cochlea.

Conclusion: Our classification based on presence of a bony modiolus in the cochlea and patency of CNC/IAC covered the all types of temporal bone malformations including cochleovestibular and CNC/IAC abnormalities and was useful to predict CI outcomes of children with temporal bone malformations.

Otology/Neurotology

Clinical Analysis of 22 Cases of Automastoidectomy

Sun Kyu Lee, MD (presenter); JiHyeon Jeung; Jae Yong Byun; Seung-Geun Yeo; Moon-Suh Park

Objective: Automastoidectomy is a destructive condition of the mastoid and middle ear cavity about which very little has been reported. 1) Know the clinical features and severities of automastoidectomy caused by chronic otitis media. 2) Evaluate the results of its surgical outcomes.

Method: Retrospective study. Patients who visited outpatient clinic and were diagnosed with COM with automastoidectomy state by temporal bone CT from January 2003 to December 2011 were included in this study. All patients underwent surgery. Their postoperative states were evaluated at the outpatient clinic at least 6 months after surgery.

Results: Two patients complained of facial palsy, 7 hearing disturbance, 10 otorrhea, and 3 otalgia. At physical examination, attic destruction and retracted tympanic membrane (TM) were observed in 12 patients, EAC destruction in 8, TM perforation in 4, and 1 had intact TM. All pathologic results were cholesteatoma. A total of 19 had canal wall down mastoidectomy, 2 had canal wall-up mastoidectomy, and 1 had subtotal petrosectomy. One had tympanoplasty type 1, 3 had columella over stapes head, 10 had interposition or columella over stapedial footplate and, 1 had tympanoplasty type 3. Overall hearing improvement rate was 9.0%. There was no recurrence.

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Conclusion: Automastoidectomy is a relatively rare condition of middle ear diseases. It could be caused by cholesteatomatous COM and non-cholesteatomatous COM. Hearing reconstructive procedures are generally not effective compared with other conditions. This study may be helpful to know the clinical features and surgical outcomes of this condition.

Otology/Neurotology

Clinical Value of VEMP in Ménière's Disease

Hye-Youn Youm, MD (presenter);
Min Beom Kim; Jeesun Choi; Ha-Young Byun;
Ga-Young Park; Won-Ho Chung

Objective: 1) To find clinical value of cervical vestibular evoked myogenic potential (VEMP) in Ménière's disease (MD). 2) To evaluate whether the VEMP results can be useful in assessing the stage of MD. 3) To evaluate the clinical effectiveness of VEMP in predicting hearing outcome.

Method: The amplitude, peak latency and interaural amplitude difference ratio (IAD ratio) was obtained using cervical VEMP. The VEMP results of MD were compared with normal subjects and MD stages were compared with IAD ratio. Finally, the hearing changes were analyzed according to their VEMP results.

Results: In clinically definite unilateral MD (n = 41), the prevalence of cervical VEMP abnormality in IAD ratio was 34.1%. Compared with healthy subjects (n = 33), the VEMP profile showed low amplitude and similar latency. The mean IAD ratio in MD was 23%, and was significantly different from healthy subjects (P = .01). As the stage increased, IAD ratio was significantly increased (P = .078). After stratification by initial hearing level, stage I and II subjects (hearing threshold, 0-40 dB) with abnormal IAD ratio showed a decrease in hearing over time compared to those with normal IAD ratio (P = .065).

Conclusion: VEMP parameters have an important clinical role in MD. Especially, IAD ratio can be used to assess the stage of MD. An abnormal IAD ratio may be used as a predictor of poor hearing outcomes in subjects with early stage MD.

Otology/Neurotology

Cochlear Implant in Far Advanced Otosclerosis

Hector E. Ruiz, MD (presenter);
Carlos Curet, MD, PhD; Maria I. Salvadores;
Carolina Romani; Luis Rubiño; Adriana Queirolo;
Gabriela Dotto

Objective: To evaluate clinical characteristics of these patients, with otosclerosis, complications,and observations of the benefits of implants to medium and long term.

Method: Thirty two adult patients, age 50 years old, 22 women and 10 men, with profound sensorineural hearing loss, and otosclerosis far advanced, were treated with multi-channel cochlear implants. All had otological examination, psychological, audiometry, and 0% of speech discriminations with-well-fitted hearing aid, CTS (4 with RMI) to determine hypodensity or morphologic changes.

Results: CTS with morphologic changes in the cochlea in 24/32 patients (Rottevel´s grading): 9 had type 2 (localized retrofenestral disease), 6 type 3 (retrofenestral diffuse). Full insertion inside the cochlea in 28/32 cases, where in 6 patients it was necessary to drill the bone in 4 to 6 mm in the tympanic basal turn. four patients had partial insertion of electrodes, 1 of them with 2 electrodes in IAC and leakage of CSF, 1 with stimulation of the facial nerve (type 3). Three patients were reinplanted. Tinnitus with sensation of resounds was observed in 5 patients. Good discriminations in speech was evidenced in 26/32 patients.

Conclusion: Patients with far advanced otosclerosis demonstrated good performance with CI in 26/32 cases. In otosclerosis type 3, 5 out of 6 patients had more difficulty in the insertion of electrodes, 1 of them with 2 electrodes within the IAC and leakage of LCR, and 5 of them with tinnitus and smaller auditory outcome.

Otology/Neurotology

Cochlear Implant Performance after Round Window Insertion

Bryan J. Kang (presenter); Derek Petti;
Ana H. Kim, MD

Objective: To compare speech performance after undergoing round window (RW) electrode insertion with those who underwent traditional cochleostomy (C) cochlear implant (CI) surgery.

Method: Study design: Prospective study. Setting: Academic cochlear implant center. Patients who met CI criteria with favorable intraoperative RW anatomy who underwent RW CI electrode insertion were included. Patients meeting CI criteria who underwent traditional cochleostomy surgery were matched for age, duration of deafness, and preoperative hearing to serve as the comparison group. Intervention: RW versus traditional C electrode insertion. Main outcome measures: Postoperative speech performance using the consonant-nucleus-consonant (CNC) speech perception tested between 3 to 24 months after CI implantation.

Results: Eighty-seven patients underwent RW insertion between 2007 to 2011. Thirty-two patients had pre- and postoperative speech perception data for analysis. Age ranged from 9.7 to 79.4 years (mean, 53.9 ± 16.1 years). A total of 18 patients who underwent C insertion were identified, ranging in age from 37.8 to 68.3 years (mean, 51.4 ± 16.1 years). No significant differences in postoperative CI speech perception scores were noted between the RW and C group at 1 year post-CI (RW group 53.6 ± 23.2% versus C group 56.2 ± 21.4%; P = .73).

Conclusion: Our study suggests that in patients with favorable RW anatomy, RW cochlear implant electrode insertion